

Ronish Bhatt

Software engineer · Toronto, Canada · open to remote

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SUMMARY

Honours BSc in Computer Science (software engineering co-op specialist, University of Toronto, 2025). I build full-stack products and run the infrastructure under them. This year: a website whose pages are signed Nostr events served from my own relay, a proof-of-work-gated community layer that needs no accounts, a Bitcoin timechain analytics tool with a from-scratch 3D renderer, a 7-million-parameter language model that runs in the browser, and independent interpretability research on small transformers.

SKILLS

languages	TypeScript, JavaScript, Python, Java, SQL, Nix
web	Node, Preact and React, NestJS, Django, Canvas 2D, Web Workers, WebAudio, service and static hosting behind Cloudflare
infrastructure	NixOS, Linux, nginx, Docker, systemd, Cloudflare Tunnel, AWS, Azure DevOps; declarative, reproducible hosts
bitcoin and nostr	Bitcoin Core and strfry relay operations, relay write-policy plugins, NIPs 01, 05, 09, 13, 17, 44, 59, 96 and 98 implemented by hand, difficulty-epoch timechain math
ml and data	PyTorch, transformers, mechanistic interpretability (weight dictionaries, superweight patching), pandas, scikit-learn, Power BI
security	responsible disclosure, API and protocol testing, Charles Proxy, threat modelling for hosted services

SELECTED WORK, 2026

ronishbhatt.com: a website made of signed events

2026

- A 22-line HTML shell; layout, stylesheet and every content block are kind 30078 events signed by the site key, fetched from a personal strfry relay with public relays as backups, verified and assembled in the browser. Subresource integrity and import-map integrity end to end.
- Content veiled with NIP-44 under a key that is the favicon itself: an 8×8 mosaic encoding 32 bytes. Server-side pre-render for first paint, a 3 KB vanilla boot script, hashed module names to survive CDN caching.
- NIP-17 gift-wrapped chat and small experiments (a mural of visitors' wraps, a word garden) all gated by NIP-13 proof of work instead of accounts, enforced by tiered relay write policy.

NOSTR ORBIT: a Nostr history on the timechain

2026

- Maps any profile's events to Bitcoin block heights using the real difficulty schedule (about five blocks of error instead of tens of thousands from a naive average), with halving-era rings, NIP-03 attestation counts and OpenTimestamps awareness.
- A voxel chain where every Bitcoin block is a cube (12×12 a day, 14 days an epoch, a coil turn a year), mempool-style epoch blocks whose contents are the notes as a treemap coloured by relay-reported engagement, a scrubbable replay with captions. Software 3D on a 2D canvas, no libraries, frames only on input.

Toronto Skating

2025 to 2026

- Live outdoor and indoor rink schedules with a Nostr community layer: chat, comments and votes without sign-ups, spam-resisted by proof of work; the relay policy plugin is being extracted as an open-source project.

Plant a Thought

2026

- A 7 M-parameter language model and a tiny inference engine running entirely in the browser, teaching it a memory during the session; built from a research codebase on context banks.

Interpretability research (independent)

2026

- Papers on weight dictionaries and superweight circuit patching in small transformers; a public demo of context-dilution attacks on GPT-2 with real logits computed in the browser.

Infrastructure

2025 to 2026

- A hardened NixOS host running Bitcoin Core, a strfry relay and the site behind Cloudflare Tunnel: declarative configuration, linted nginx, a zero-SSH operations dashboard, write policy as the security boundary.

OPEN SOURCE

- **pandas** (2023): implemented `escape="latex-math"` for LaTeX table rendering so math mode survives while unsafe characters are still escaped (issue #50040), with unit tests across edge cases; co-authored the test plan for the `merge(..., combine=...)` enhancement (#16070).

EXPERIENCE

Financial Technology Risk Analyst (co-op)

May to Aug 2021

Office of the Superintendent of Financial Institutions, Toronto

- Built the Power BI and Excel dashboard for 320+ cyber incidents across regulated institutions; cut redundant categories by a fifth.
- Built an NLP pipeline (sentiment and topic modelling) that surfaced emerging risks and led to three new controls.
- Automated MISP deployment on Ubuntu, taking setup from three days to three hours.

Web Developer (co-op)

Jan to Apr 2022

Media Matters, Toronto

- Shipped six responsive React pages that lifted average session duration by about 15 percent.

- Automated the HubSpot hand-off, cutting lead-routing time by a quarter; scraped 1,000+ dealership sites to feed a market-analysis dashboard.

Security testing (independent)

2018 to present

- Responsibly disclosed a double-spend-class race condition at a major crypto platform (under NDA), a promotional-credit accrual loophole in a large e-commerce mini-app, and a pack-opening exploit in a Web3 card game; vendors patched.

Administrative Assistant

Apr to Aug 2025

Non-Residential Nepali Association, Toronto

- Supported AGM financial statements, election logistics and payroll for a national organization.

EDUCATION

Honours Bachelor of Science, Computer Science

2019 to 2025

University of Toronto Scarborough · Specialist in Software Engineering, co-op · University of Toronto Scholar Award